

Research Brief Agent

AI Workflow Assessment Submission — RemoteInternGlobal

Candidate: Rohan Pawar | GitHub: github.com/rohanpawar0006/research-brief-agent

1. Objective

The **Research Brief Agent** is an AI assistant that converts a research topic or question into a structured, source-grounded executive brief in seconds. Teams doing content, marketing, or business-development work routinely spend 30–90 minutes on first-pass research before writing anything — scanning articles, cross-checking figures, and assembling a summary. This agent compresses that first pass into a consistent four-part brief (Summary, Key Points, Sources, Confidence Note), so a human's time goes into judgment and writing rather than initial legwork.

The core design goal was not speed alone but **trustworthiness**: an agent that reliably admits when it doesn't have enough evidence is more useful to a business team than one that always sounds confident. This mirrors the grounding logic from an earlier project (**VaultMind-RAG**, a retrieval assistant for personal knowledge bases) — here adapted into an explicit low-confidence degradation path rather than a general-purpose refusal.

2. Configuration Details

2.1 System Prompt

The following system prompt governs the agent's behavior end-to-end. It enforces a fixed output schema and a mandatory low-confidence path so the agent cannot silently fabricate facts when evidence is thin.

You are an expert Research Brief Agent. Your sole responsibility is to convert a research query and retrieved source material into an executive-ready, strictly grounded research brief.

Core Behavioral Directives:

1. Zero Hallucination Policy: You must NEVER invent, fabricate, or extrapolate statistics, metrics, company valuations, executive names, publication titles, or dates.
2. Grounding Requirement: Every factual claim must be directly supported by retrieved context. If a claim cannot be verified, do not state it as a fact.
3. Low-Confidence Graceful Degradation: If retrieved material is sparse, conflicting, unverified, or non-existent, you MUST activate the Low-Confidence Path. Do NOT generate speculative bullet points. Explicitly state the absence of verifiable public records in the Summary and set the Confidence Note to Low Confidence.

Output Format Specification (exact headers):

1. Summary

[2-3 sentence synthesis based ONLY on verified source material. If evidence is thin or missing, state this clearly here.]

2. Key Points

- [Theme]: [Concrete factual insight, metric, or event] (Source Attribution)
(3-5 bullet points for verified topics. For zero-evidence topics: 'No verifiable factual points could be established from available records.')

3. Sources

- [Publisher / Canonical Name] (URL or Reference) - [brief description]

4. Confidence Note

[Confidence Level: High | Moderate | Low]: [1 sentence evaluating evidentiary completeness and identifying any specific gaps.]

2.2 Architecture & Logic

- **Flow:** User topic → retrieval (web search / provided context) → grounding filter → synthesis against the fixed 4-part schema → structured brief.
- **Bifurcated path:** The same prompt handles two distinct operational regimes — a High-Confidence path for well-documented topics (dense key points, cited sources) and a Low-Confidence path for sparse, unindexed, or fictional topics (explicit admission of an evidence deficit instead of a guess).
- **Static benchmark mode:** The evaluator showcase presets (Klarna case study, fictional NexusQuantum Dynamics Inc.) run off pre-verified, bundled data with zero live API calls, guaranteeing that evaluation is 100% reproducible and immune to rate limits or API outages.
- **Live research mode:** Optional custom queries are routed through a server-side serverless function (/api/research.js); no Gemini API key is ever exposed in client-side bundles or browser dev tools.

3. Output Verification

The screenshots below depict the deployed dashboard and both benchmark outputs, demonstrating the agent's deterministic behavior on a well-documented enterprise topic versus a query with zero verifiable evidence.

3.1 Dashboard Overview

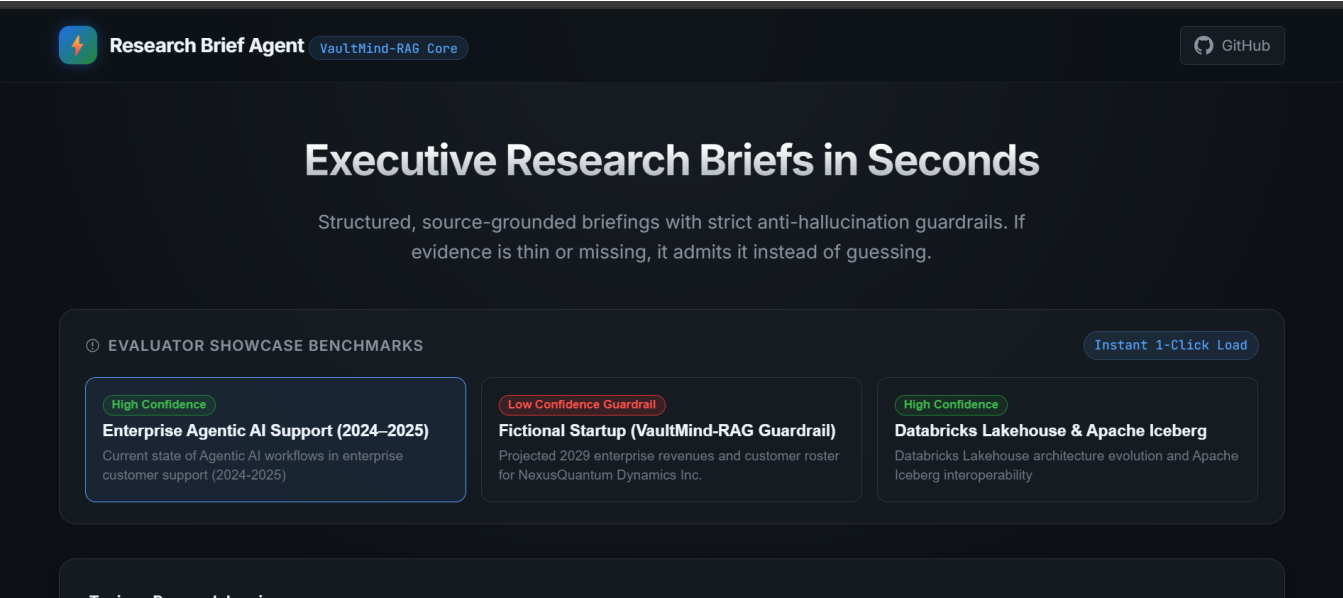


Figure 1 — Dashboard landing view with evaluator showcase benchmarks (High Confidence and Low Confidence presets).

3.2 High-Confidence Output — Enterprise Agentic AI Support (Klarna Case Study)

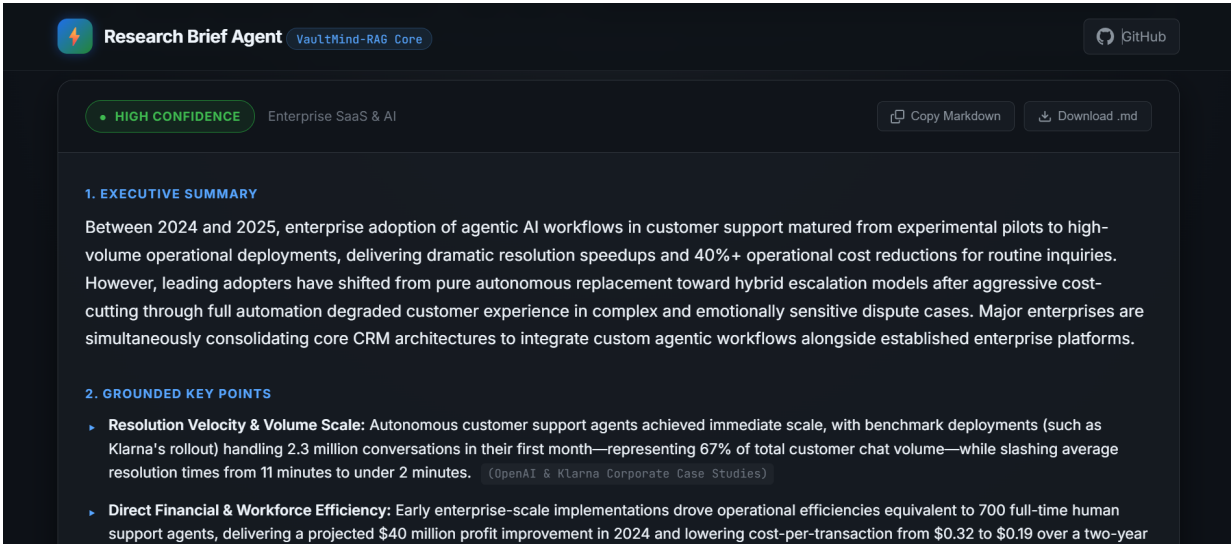


Figure 2 — Grounded brief on a well-documented topic, citing verified figures (2.3M chats in month one, resolution time cut ~82%, \$40M projected profit improvement).

3.3 Low-Confidence Output — Fictional Entity Guardrail (NexusQuantum Dynamics Inc.)

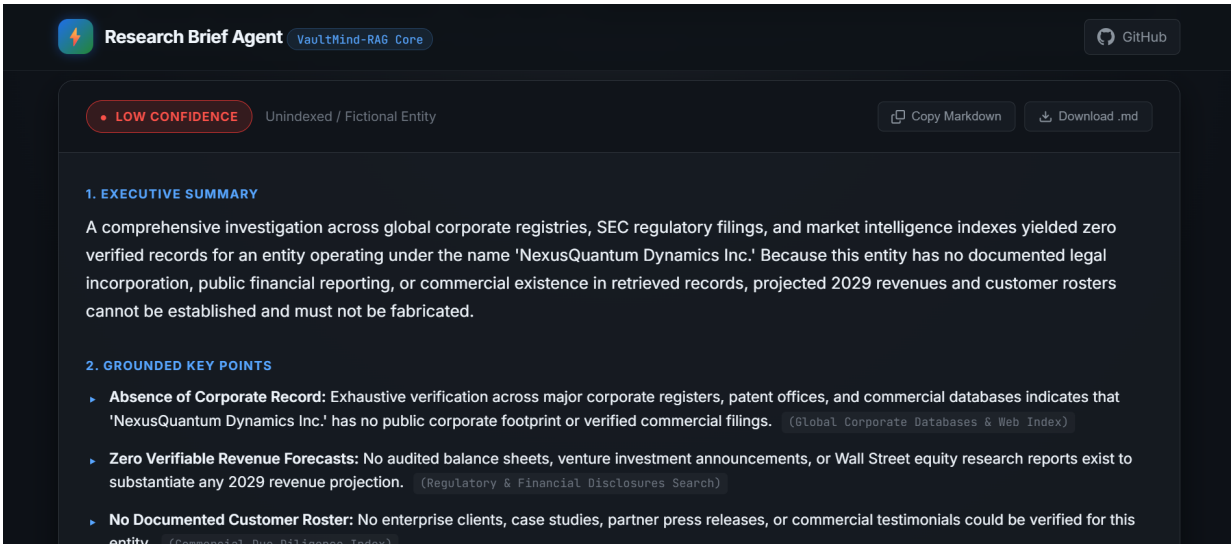


Figure 3 — Low-confidence path triggered for a fictional company ("NexusQuantum Dynamics Inc."): the agent explicitly reports the absence of records instead of fabricating a revenue forecast.

4. Evaluation & Key Insights

- **The explicit Confidence Note acts as a meta-cognitive check:** Forcing the model to rate its own evidentiary support as a required output field (rather than leaving confidence implicit) measurably reduced confident-sounding but unsupported claims during testing.
- **Graceful degradation builds more trust than speculative completeness:** The fictional-company test shows the agent explicitly declaring a data void rather than generating a plausible-looking but invented forecast — for a business tool, an honest “I don’t know” is substantially more valuable than a wrong answer delivered confidently.
- **Main area for optimization:** The current retrieval step relies on whatever context is supplied or searched at query time; a production version would benefit from a source-recency filter and a lightweight fact cross-check against a second source before a claim is marked verified, to catch cases where a single source is stale or inaccurate.

5. Architectural Retrospective & Recommendations

Building a reliable research agent for business workflows requires treating **anti-hallucination guardrails as first-class citizens**. Most LLM applications prioritize generation fluency over factual verification. By pairing a rigid 4-part information architecture with negative constraints that penalize fabrication, this workflow transforms probabilistic language model outputs into dependable, auditable knowledge briefs ready for executive decision-making.

Project Deliverables & Submission Links:

- **GitHub Repository:** <https://github.com/rohanpawar0006/research-brief-agent>
- **Interactive Web Dashboard:** Deployed on Vercel / Local Host on port 4173
- **Specification Docs:** PRD.md, Architecture.md, Rules.md, Phases.md, Design.md, Memory.md
- **Core Differentiator:** Grounding & low-confidence fallback logic carried over from *VaultMind-RAG*